⚙️ Fixed Parameters (무차원 수 입력)

D_eo*

0.802 − +

D_bi*

0.700 − +

D_oi*

1.200 − +

Pitch p*

0.700 − +

Starts N_s*

1.200 − +

Heat Exchanger Parametric Analysis Dashboard

📌 Current Geometric Parameters (계산된 형상 치수)

D_eo (실제값)
36.09 mm
↑ D_eo* = 0.802

D_bi (실제값)
24.50 mm
↑ D_bi* = 0.700

D_oi (실제값)
54.00 mm
↑ D_oi* = 1.200

Pitch p (실제값)
17.50 mm
↑ p* = 0.700

Starts N_s (실제값)
4.80
↑ N_s* = 1.200

Tube Inner Dia (D_vi)
28.13 mm

Tube Outer Dia (D_vo)
29.13 mm

Hydraulic Dia (D_hyd)
24.87 mm

Rib Height (e)
0.05 mm

Helix Angle (θ)
47.45°

📊 Performance Results

Total Length (L)
59.549 m
↑ L* = 0.866

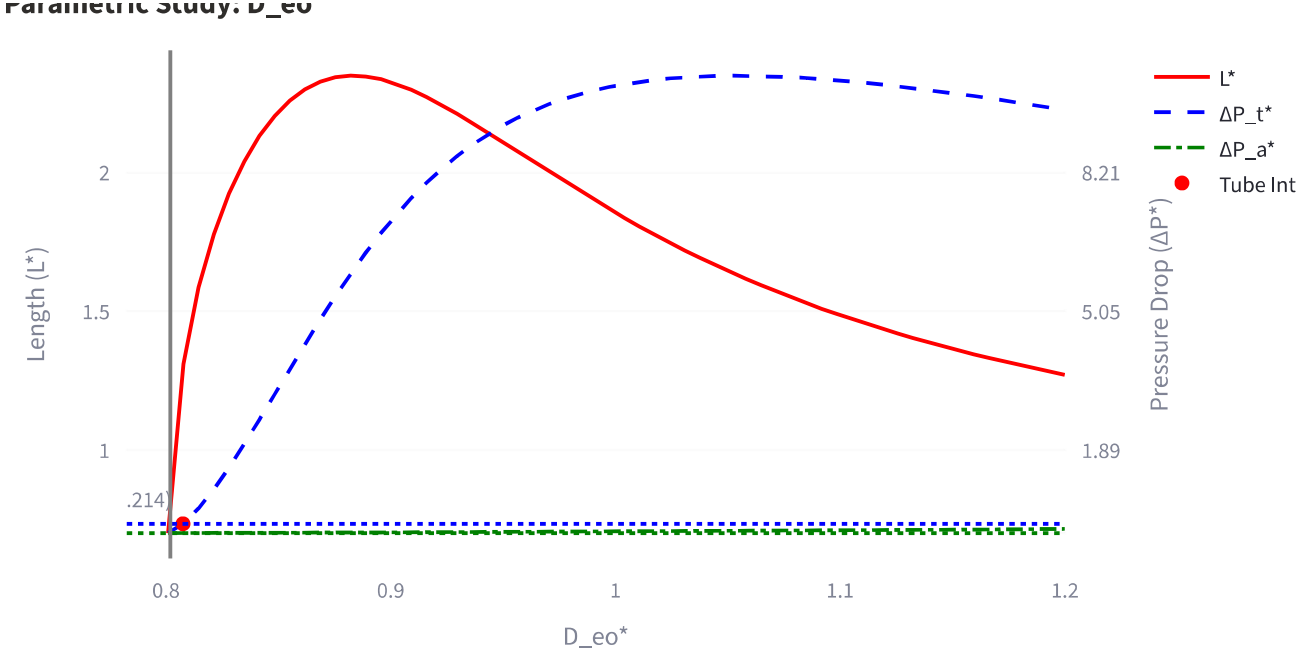
Tube PD (ΔP_t)
6.5 kPa
↑ ΔP_t* = 0.028

Annular PD (ΔP_a)
71.5 kPa
↑ ΔP_a* = 0.005

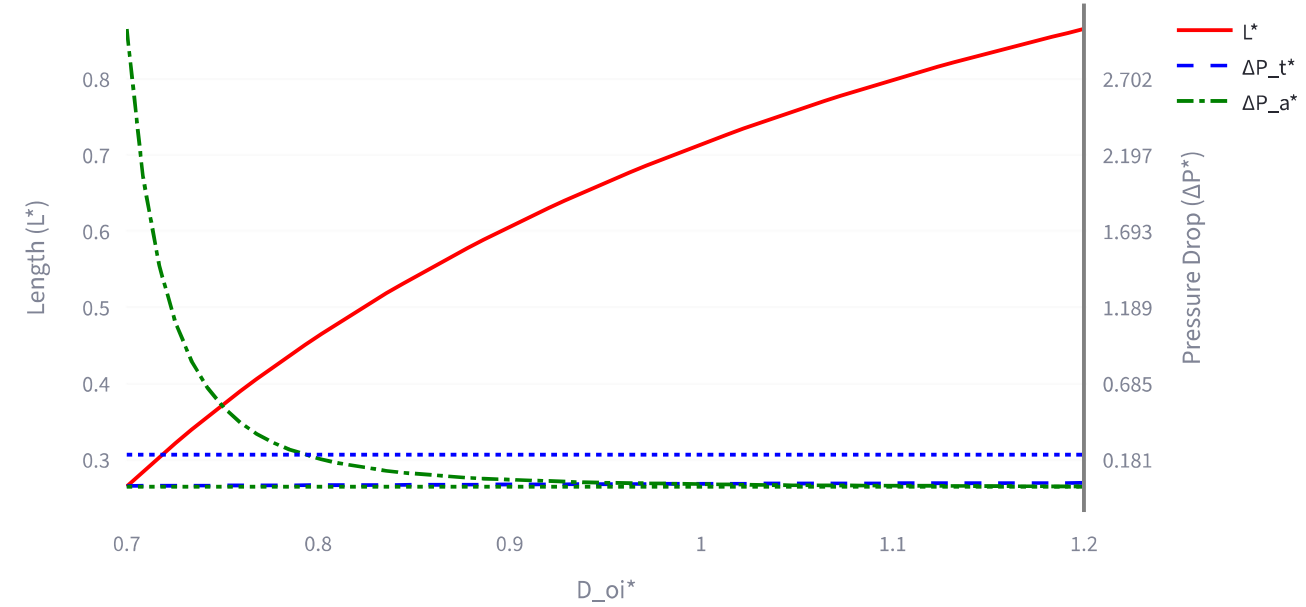
Cold Outlet Temp (Tc_out)
78.18 °C

Heat Duty (Q)
195.4 kW

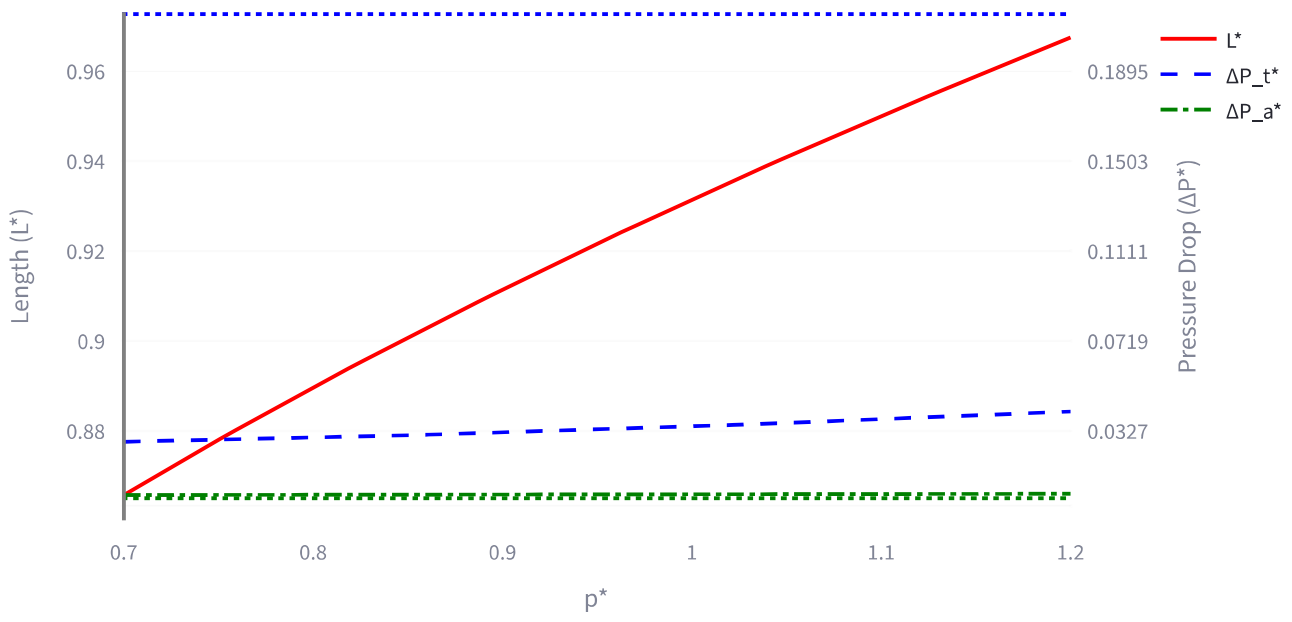
Parametric Study: D_eo*



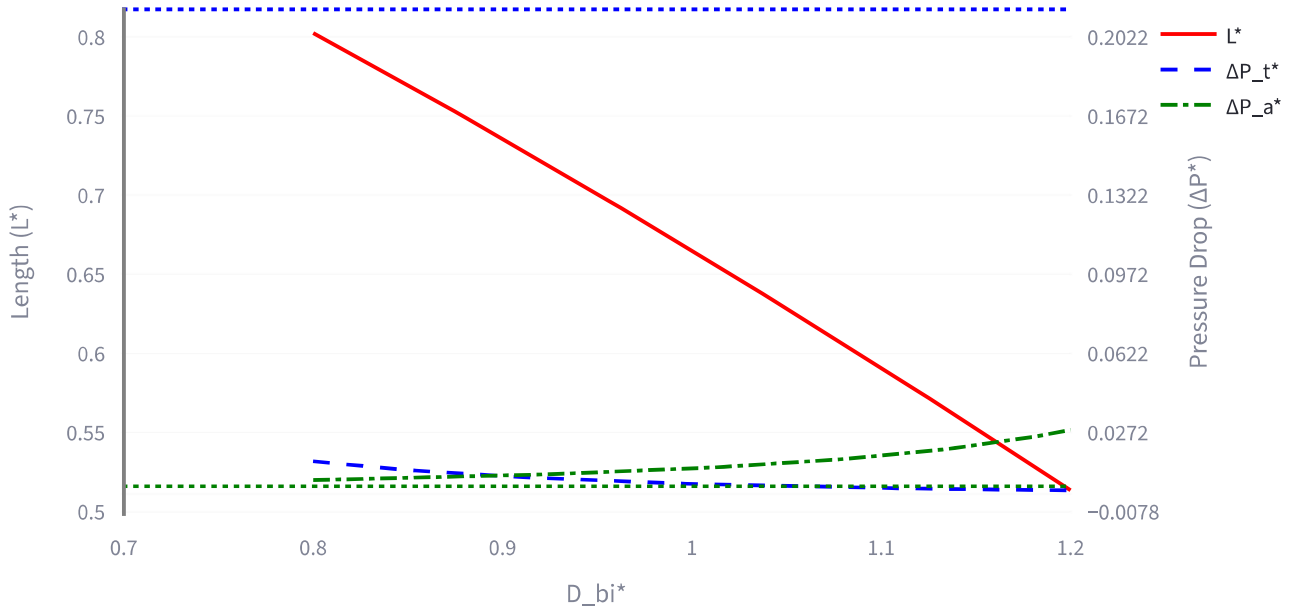
Parametric Study: D_oi*



Parametric Study: p*



Parametric Study: D_bi*



Parametric Study: N_s*

